CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: November 2019 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Differentiate between Bridges and Repeaters.

Differences: Repeaters operate at the Physical Layer (Layer 1) to regenerate electrical signals without looking at frames. Bridges operate at the Data Link Layer (Layer 2) to inspect MAC addresses and filter traffic between segments.

Q2 (3M) What are the services provided by the Transport Layer?

Services: Connection-oriented delivery (TCP) vs connectionless datagram delivery (UDP), port-based process addressing, segmentation/reassembly, end-to-end flow control, and error recovery.

Q3 (3M) Define Modulation and Demodulation.

Explanation: Modulation transforms digital bitstreams into analog signals for transmission across physical media. Demodulation recovers original digital bits from incoming modulated analog waves.

Q4 (3M) What is Network Topology? Name two topologies with advantages.

Topologies: Star topology (central hub, easy fault isolation) and Mesh topology (point-to-point dedicated links, highest fault tolerance).

Q5 (3M) What is Shannon Capacity Theorem?

Theorem: C = B \log_2(1 + SNR) gives the theoretical upper bound on data transmission speed in bits per second over a noisy channel.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain Dijkstra's Algorithm with an example network.

Dijkstra's Algorithm: Finds single-source shortest path to all destinations by maintaining tentative distance labels and iteratively selecting the minimum-cost unvisited node.

Q7 (7.5M) Explain Distance Vector Routing and the Count-to-Infinity problem.

Distance Vector Routing: Uses Bellman-Ford algorithm; slow reaction to link failures causes loops (count-to-infinity), resolved via split horizon.

Q8 (7.5M) Explain Sliding Window Protocol for Flow Control.

Sliding Window: Permits sending multiple frames within a window without waiting for immediate individual acknowledgments.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) Explain ISO/OSI Reference Model in detail with all seven layers.

OSI Architecture: Full 7-layer functional breakdown and data encapsulation process.

Q10 (15M) **Explain Guided Media: Twisted Pair Cable, Coaxial Cable, and Fiber Optic Cable.**

Transmission Media: Construction, signal propagation, attenuation characteristics, and network cabling standards (Cat5e, Cat6, Single-mode/Multi-mode fiber).

Q11 (15M) **Explain Error Detection using CRC (Cyclic Redundancy Check) with numerical problem.**

CRC Problem: Generator polynomial division, bit stuffing, and frame validation on receiver side.

Q12 (15M) **Explain Circuit Switching, Packet Switching, and Message Switching with comparison.**

Switching Techniques: Comprehensive comparison of latency, bandwidth efficiency, and modern telecom implementations.

Q13 (15M) **Explain Congestion Control in Computer Networks: Leaky Bucket and Token Bucket Algorithms.**

Traffic Shaping: Deep dive into burst mitigation, token accumulation, and packet scheduling.
